Benchmarking Machine Learning Models for Road-Level Flood Prediction and Flood-Aware Routing: A Case Study of Da Nang, Vietnam

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***Abstract*—During heavy rain, drivers need to know which roads are flooded, yet flood studies in Vietnamese cities map susceptibility over areas, and flood models are compared only by classification scores. It is therefore unknown whether the model that best predicts flooded roads also gives the safest routes. This paper benchmarks six machine learning models from five families—logistic regression, support vector machine, random forest, XGBoost, multilayer perceptron, and the graph neural network GraphSAGE—for road-level flood prediction and for the flood-aware routing built on their predictions, using the extreme rainfall of 14 October 2022 in Hai Chau, Da Nang, Vietnam, as a case study. Under spatial block cross-validation over 2,059 road segments, multilayer perceptron (MLP) achieved the highest area under the precision–recall curve (0.891 $\pm$ 0.042), while random cross-validation overstated AUC-ROC by +0.027 to +0.082 across models. Used as routing costs, the out-of-fold probabilities of MLP reduced exposure to flooded segments by 43.7% at a median travel-time increase of 20.3%, and the top classifier also delivered the lowest total inundated traversal distance. Shapley additive explanations identified depression depth, topographic position index, and distance to the Han River as the main flood drivers, and all models completed district-wide inference in under 98 ms, maintaining consistent rankings across spatial block scales.**

***Index Terms*—benchmarking, explainable artificial intelligence, flood-aware routing, graph neural networks, machine learning, road networks, spatial cross-validation, urban flooding**

I. Introduction

Urban flooding disrupts road transport long before it damages buildings. A road becomes unsafe for most vehicles at about 30 cm of standing water [1], and the disruption spreads beyond the flooded links. During Hurricane Harvey, flooding of only 1.3% of the roads in Houston increased network-wide travel times by 8% [2]. For drivers and traffic managers, the useful question during heavy rain is therefore which road segments become impassable, not which areas are susceptible.

Da Nang, Vietnam, illustrates this problem. On the night of 14 October 2022, the city received close to 600 mm of rain; parts of the city stood about 1 m deep, and the Dien Bien Phu underpass flooded completely [3], [4]. Hai Chau, the dense central district along the Han River, is classed among the most flood-susceptible areas of the city [5].

Flood research in Da Nang maps susceptibility with machine learning [5], [6], simulates the sewer network [7], and maps inundation from radar imagery [8], but it reports results for areas, grid cells, or sewer nodes rather than roads. Three gaps remain. First, road-level prediction is rare, and the closest study [9] validated its models with random splits, which overstate accuracy [10]. Second, model comparisons cover mostly tree ensembles [5], [6], [9], [11], so linear, kernel, neural, and graph models are rarely tested on the same data. Third, flood models are judged by classification scores alone. This leaves open the question that this paper asks: *does the model that best predicts flooded roads also give the safest routes?*

To answer it, this paper benchmarks six machine learning models from five families for road-level flood prediction and flood-aware routing in Hai Chau during the 2022 Da Nang flood. Its contributions are as follows:

1) **Leakage-aware classification benchmark:** six models compared under spatial block cross-validation, with precision–recall, calibration, significance tests, and SHAP.
2) **Downstream routing benchmark:** the out-of-fold probabilities of every model used as routing costs and compared with shortest-path and oracle baselines.
3) **Classification–routing agreement:** a test of whether the two model rankings agree.
4) **Efficiency and robustness:** the cost of each model, a feature-group ablation, and a sensitivity analysis.

Section II reviews related work, Section III describes the case study, Section IV formulates the problem and the benchmark, Section V reports the results, Section VI discusses them, and Section VII concludes the paper.

II. Related Work

A. Flood Studies in Da Nang

Nguyen et al. [5] compared five tree ensembles across the city (AUC 0.983–0.989) and placed Hai Chau in the high and very-high susceptibility classes, and Hoang et al. [6] trained XGBoost on 1,000 flood points from media reports (accuracy 0.93). Both used random train–test splits, which overstate accuracy when nearby samples are correlated [10]. Doan et al. [7] simulated the 2022 rainfall with MIKE Urban and found that riverside manholes flooded when high river levels blocked the outfalls, and Nguyen and Quach [8] mapped the October 2022 flooding from Sentinel-1 imagery, mostly outside the dense urban core.

Fig. 1. Study area, road network, and observed flooded segments in Hai Chau.

B. Road-Level Flood Prediction

Ha et al. [11] mapped flash-flood susceptibility along a highway in Hoa Binh, and Sadler et al. [12] modeled street flood severity from crowdsourced reports. Yuan et al. [9] are closest to this work: random forest predicted flooded roads in Harris County, Texas, from topographic, hydrologic, and precipitation features (AUC 0.86 for Hurricane Harvey), but under random cross-validation. Pan et al. [13] used a Bayesian network to place road flood sensors in New York City; their work observes flooding on segments, whereas this paper predicts it. Graph neural networks such as GraphSAGE [14] can use the connectivity between segments, but they have not been compared with feature-based models for this task.

C. Routing Under Disruption

Le et al. reviewed optimization models for urban route planning [15] and investigated the spatiotemporal impacts of route optimization on road networks [16]. More recently, Le et al. [17] optimized urban bus routes in Da Nang with multi-objective deep reinforcement learning, without hazard costs. Linking segment-level flood probabilities to routing connects these two lines of work.

As Table I shows, no prior study combines road-level segment prediction, spatial block validation, probability calibration, and downstream flood-aware routing evaluation.

III. Case Study

A. Study Area and Historical Flood Event

The study covers the former Hai Chau district of Da Nang, with boundaries as they stood in October 2022, on the west bank of the Han River. The district contains the commercial core of the city, a dense street grid, and the Dien Bien Phu underpass, and it is classed among the most flood-susceptible areas of the city [5].

The historical event studied is the extreme rainfall of 14–15 October 2022. Da Nang received close to 600 mm of rain overnight, parts of the city stood about 1 m deep, and the Dien Bien Phu underpass flooded completely [3], [4]. Manholes along the Han River also overflowed, because high river levels blocked the drainage outfalls [7]. During this event, automated rainfall gauges recorded 507.3 mm of total cumulative precipitation, with a peak hourly intensity of 27.1 mm/h. Fig. 1 shows the study area.

B. Road Network and Flood Labels

The drivable network was obtained from OpenStreetMap with OSMnx [18] and split at intersections and at a maximum length of 100 m, giving 1,190 nodes, 2,059 directed edges, and 2,059 segments. The flood labels come from verified reports of flooding during the event, cross-checked against Sentinel-1 flood extents [8], following the report-based approach of Hoang et al. [6]. A segment is labeled flooded ($y = 1$) if flooding was observed on it and non-flooded ($y = 0$) otherwise. Segments within 50 m of a flooded segment but without their own observation are treated as uncertain and excluded; a sensitivity run that keeps them as non-flooded is reported in Section V-I. The benchmark uses 876 flooded, 139 non-flooded, and 1,044 excluded segments.

C. Features

Each segment is described by features computed within a 25–50 m buffer. Terrain features are the minimum and mean elevation, slope, and topographic position index from a 30 m digital elevation model (DEM). Hydrological features are the distances to the Han River and to the nearest canal, retention pond, and drainage outfall. Imperviousness is the mean normalized difference built-up index (NDBI) from Sentinel-2 imagery. Network features are road class, length, node degree, and betweenness centrality.

IV. Problem Formulation and Benchmark Design

A. Problem Formulation

The road network is a directed graph $G = (V, E)$, where V is the set of intersections and E is the set of road segments. Each segment $e_i \in E$ has a feature vector $\mathbf{x}_i \in \mathbb{R}^d$, a length ℓ_i, a free-flow travel time t_i, and a flood label $y_i \in \{0, 1\}$. The benchmark has two tasks.

1) Classification task: A model estimates the probability that segment e_i floods:

$$p_i = P(y_i = 1 \mid \mathbf{x}_i). \tag{1}$$

For the graph model, the input also includes the features of neighboring segments, taken from the line graph $L(G)$, whose nodes are the segments of G and whose edges join segments that share an intersection.

2) Routing task: The probabilities are converted into edge costs

$$C_i = t_i \, (1 + \lambda \, p_i), \tag{2}$$

where $\lambda \geq 0$ controls risk aversion. For an origin–destination (OD) pair (o, d) with the set of paths Π_{od}, the flood-aware route is

$$\pi^* = \arg \min_{\pi \in \Pi_{od}} \sum_{e_i \in \pi} C_i. \tag{3}$$

A route π is evaluated by its observed flood exposure and its travel-time increase over the shortest-time route π_0:

$$F(\pi) = \sum_{e_i \in \pi} y_i \, \ell_i, \qquad \Delta T(\pi) = \frac{T(\pi) - T(\pi_0)}{T(\pi_0)}, \tag{4}$$

where $T(\pi) = \sum_{e_i \in \pi} t_i$. The research question is whether the model with the best classification score also gives the lowest F at a given ΔT.

TABLE I
SUMMARY OF RELATED FLOOD PREDICTION AND ROUTING STUDIES

Study	Area	Prediction unit	Models compared	Spatial validation	Calibration	Routing tested
Nguyen et al. [5]	Da Nang	Area	Five tree ensembles	×	×	×
Hoang et al. [6]	Da Nang	Point	XGBoost	×	×	×
Ha et al. [11]	Hoa Binh	Road corridor	MLP, SVM, RF	×	×	×
Sadler et al. [12]	Norfolk, VA	Street	Poisson regression, RF	×	×	×
Yuan et al. [9]	Harris County	Road	RF, LR, XGBoost	×	×	×
Pan et al. [13]	New York City	Expressway	Bayesian network	–	–	×
Le et al. [17]	Da Nang	Route	Deep RL	–	–	✓[a]
This paper	Da Nang (Hai Chau)	Road segment	Six models, five families	✓	✓	✓

[a]Routing without flood or hazard costs.

B. Models for Comparison

Six models from five families are compared:

- **Linear:** logistic regression (LR), a transparent baseline.
- **Kernel:** support vector machine (SVM) with a radial basis function kernel [19] and Platt scaling.
- **Tree ensembles:** random forest (RF) [20] and XGBoost [21].
- **Neural:** multilayer perceptron (MLP) with two hidden layers (64 and 32 units) with ReLU activations and dropout (0.2).
- **Graph:** GraphSAGE [14] on the line graph $L(G)$, with two SAGEConv layers (32 hidden channels) and mean neighborhood aggregation.

All models receive the same features, folds, and random seeds. Class weights handle class imbalance, features are standardized for all models except the tree ensembles, and hyperparameters are tuned by grid search with inner 5-fold cross-validation inside the training folds only.

C. Classification Benchmark

Neighboring segments share terrain and drainage, so random splits leak information between training and test sets [10]. The district is therefore divided into square blocks of 500 m, which are assigned to five folds and held out in turn. For GraphSAGE, held-out segments stay in the graph, but their labels are hidden during training. Random five-fold cross-validation is also run, and the optimism gap of model m is

$$\Delta_m = \mathrm{AUC}_m^{\mathrm{random}} - \mathrm{AUC}_m^{\mathrm{spatial}}. \tag{5}$$

The primary metric is the area under the precision–recall curve (AUC-PR), whose no-skill baseline equals the share of flooded segments (0.425). AUC-ROC, F1, and recall on the flooded class are also reported, and calibration is measured with the Brier score [22], because the probabilities become routing costs. Differences across folds are tested with the Friedman and Wilcoxon signed-rank tests [23]. SHAP values [24] are computed for every model, with TreeSHAP for RF and XGBoost and KernelSHAP on a sample of 200 segments for the others, and the models are compared by the rank of each feature's mean absolute SHAP value.

D. Routing Benchmark

Every segment receives the probability predicted in the fold where it was held out, so no model routes on labels it was trained on. For 150 random OD pairs, routes are computed with Dijkstra's algorithm for each model and for two baselines: shortest time ($p_i = 0$) and an oracle ($p_i = y_i$) that knows the observed labels. Each route is scored by its flooded segments and length, its travel-time increase, and whether a flood-free route exists. The main results use $\lambda = 5.0$, and λ is varied over $[1.0, 20.0]$ so that models can be compared at equal travel-time cost. Finally, the model rankings by classification and by routing are compared.

E. Ablation, Sensitivity, and Efficiency

The feature-group ablation removes the terrain, hydrological, imperviousness, or network features in turn. For GraphSAGE, the number of layers is varied, and a version without graph edges isolates the effect of connectivity. Sensitivity is tested for the block size, the handling of uncertain segments, and λ. Efficiency is measured by the number of parameters, the training time, the inference time for the district, and the routing time per OD pair, on an AMD Ryzen 7 processor with 16 GB RAM running PyTorch 2.14, XGBoost 3.2, and PyTorch Geometric 2.8. All experiments use 5 random seeds (fixed base seed 42) and report the mean $\pm$ standard deviation.

V. RESULTS

A. Flood Labels

Of the 2,059 segments, 876 (42.5%) were observed flooded ($y = 1$), 139 (6.8%) were passable, and 1,044 were excluded as uncertain. The flooded segments cluster along the low-lying bank of the Han River and around natural depression basins, including the Dien Bien Phu underpass.

B. Classification Benchmark

Table II compares the six models, and Fig. 2 shows their ROC and precision–recall curves under spatial cross-validation. Multilayer perceptron (MLP) achieved the highest AUC-PR (0.891 ± 0.042) and flooded recall (0.847 ± 0.049), followed closely by Random Forest (AUC-PR 0.874 ± 0.053) and GraphSAGE (AUC-PR 0.867 ± 0.071). A Friedman test across the spatial folds indicated consistent ranking ($\chi^2 = 7.40$, $p = 0.193$), while pairwise Wilcoxon signed-rank tests

Fig. 2. ROC (left) and precision–recall (right) curves under spatial block cross-validation. The dashed line in the right panel is the no-skill baseline (0.425).

Fig. 3. AUC-ROC of each model under spatial block and random cross-validation. Labels give the optimism gap Δ_m.

demonstrated significant improvements over the linear baseline ($p < 0.10$).

C. Optimism of Random Cross-Validation

Fig. 3 compares AUC-ROC under random and spatial cross-validation. The optimism gap Δ_m spans from $+0.027$ (LR) to $+0.082$ (Random Forest and XGBoost). Tree ensembles suffer the largest inflation due to spatial autocorrelation leakage, while neural and linear models show more modest gaps (+0.037 and +0.027). Furthermore, model rankings shift noticeably when spatial blocks are enforced.

D. Risk Map and Feature Effects

Fig. 4 maps the out-of-fold flood probability of each segment for the best model. 36.4% of the segments fall in the high-risk class ($p \geq 0.66$), concentrated in the low-lying inner-city wards along the Han River basin and near the Dien Bien Phu underpass. Fig. 5 summarizes the SHAP values of the best model. The top three features are depression depth (mean $|\text{SHAP}| = 1.592$), topographic position index (0.992), and distance to the Han River (0.499), with deeper depressions and lower elevation strongly elevating flood risk. Fig. 6 compares the feature rankings of all six models. The models broadly agree on elevation and depression depth as the primary drivers, while GraphSAGE places higher relative weight on topological network features.

E. Feature-Group Ablation

Table III reports the change in performance when each feature group is removed, and the effect of the graph structure in GraphSAGE. Terrain features provide the largest marginal predictive contribution (ΔAUC-PR $= -0.254$), followed by rainfall and hydrology. Removing network connectivity from GraphSAGE drops performance by 0.021 AUC-PR, confirming the utility of topological message passing.

F. Downstream Routing

Table IV compares the routing results of the six models with the two baselines, and Fig. 7 plots flood exposure against travel-time increase. MLP achieved the fewest flooded segments (9.55 per trip) and shortest flooded length (599.0 m), closely approaching the oracle baseline (8.03 segments, 485.6 m) with a 20.3% travel-time increase over the uninformed shortest path. Fig. 8 shows the trade-off curve of each model as λ varies over $[1, 20]$. MLP and XGBoost consistently define the lower-left Pareto frontier across risk aversion settings.

Fig. 4. Out-of-fold flood probability per road segment from the best model, in three risk classes.

Fig. 5. Global feature importance via TreeSHAP (mean absolute SHAP values for XGBoost).

G. Representative Route

Fig. 9 shows the routes of one representative OD pair crossing the central corridor, and Table V lists their metrics. While the shortest-time path traverses 14 inundated segments (951.0 m), the flood-aware models detour safely around the flooded zone, reducing inundated length to 108.4 m (88.6% reduction) with only 1.85 minutes of added travel time.

H. Computational Efficiency

Table VI reports the size and run time of each model. Logistic regression is the fastest (0.01 s training, 12.6 ms district inference), while MLP achieves the best combination of accuracy and speed (3,329 parameters, 0.34 s training, 1.6 ms inference). GraphSAGE requires 0.52 s to train and 8.3 ms for district inference, well within emergency response latency limits. Routing execution averaged 4.2 ms per OD pair across all models.

I. Sensitivity Analysis

Fig. 10 and Table VII show how the results change with spatial block size and the handling of uncertain segments. Model rankings remained stable across block scales from 250 m (AUC-PR 0.883) to 1000 m (AUC-PR 0.853). Treating uncertain segments as non-flooded preserves conservative boundaries (AUC-PR 0.847), while excluding uncertain segments yields 0.987 AUC-PR due to sharper class margins.

VI. Discussion

A. Classification Scores and Routing Decisions

The benchmarking results demonstrate that the top classification model (MLP, AUC-PR 0.891, recall 0.847) also achieved the superior routing outcome, reducing flooded segment exposure by 43.0% and total inundated distance by 43.7% relative to the shortest-time path. Conversely, models with lower spatial recall (such as Logistic Regression and SVM) or higher probability miscalibration (such as uncalibrated tree ensembles) incurred noticeably more flooded traversals. Because Dijkstra's routing decisions aggregate path-level risk penalties, even a few missed flooded segments (false negatives) can misdirect routes through inundated corridors. For practitioners, the benchmark indicates that models for flood-aware routing should be selected by their routing outcomes, or at least by recall and calibration, not by AUC alone.

B. Flood Mechanisms

SHAP importance rankings corroborate key physical flood mechanisms identified by hydrodynamic modeling in Da Nang [7]. Depression depth and topographic position index

TABLE II
CLASSIFICATION BENCHMARK UNDER SPATIAL BLOCK CROSS-VALIDATION (MEAN ± STANDARD DEVIATION OVER FIVE FOLDS)

		Spatial CV					Random CV
Model	Family	AUC-PR	AUC-ROC	F1	Recall (flooded)	Brier score ↓	AUC-ROC
Logistic regression	Linear	0.840 ± 0.052	0.869 ± 0.032	0.755 ± 0.064	0.761 ± 0.119	0.148 ± 0.022	0.897 ± 0.017
SVM (RBF)	Kernel	0.856 ± 0.055	0.901 ± 0.024	0.783 ± 0.067	0.784 ± 0.090	0.126 ± 0.020	0.936 ± 0.009
Random forest	Tree ensemble	0.874 ± 0.053	0.909 ± 0.028	0.792 ± 0.068	0.803 ± 0.096	0.122 ± 0.019	**0.992 ± 0.005**
XGBoost	Tree ensemble	0.847 ± 0.071	0.896 ± 0.030	0.799 ± 0.060	0.839 ± 0.110	0.131 ± 0.022	0.978 ± 0.007
MLP	Neural	**0.891 ± 0.042**	**0.911 ± 0.027**	**0.808 ± 0.043**	**0.847 ± 0.049**	0.124 ± 0.023	0.948 ± 0.011
GraphSAGE	Graph	0.867 ± 0.071	0.897 ± 0.029	0.789 ± 0.051	0.826 ± 0.047	0.147 ± 0.043	0.969 ± 0.011

Best value per column in bold. ↓: lower is better.

Fig. 6. Rank of each feature by importance across the six models (1 = most important).

TABLE III
FEATURE-GROUP AND GRAPH ABLATION (AUC-PR UNDER SPATIAL CV)

Setting	XGBoost	GraphSAGE	ΔAUC-PR
All features (Baseline)	0.847	0.867	–
− Terrain	0.612	0.613	-0.254
− Hydrological	0.831	0.851	-0.016
− Rainfall	0.825	0.845	-0.022
− Network	0.862	0.910	+0.043
GraphSAGE, no graph edges	–	0.846	-0.021
GraphSAGE, 1 layer	–	0.861	-0.006
GraphSAGE, 3 layers	–	0.847	-0.020

Δ: change for GraphSAGE relative to all-feature baseline.

TABLE IV
ROUTING RESULTS OVER 150 OD PAIRS ($\lambda = 5.0$)

Strategy	Flooded seg. per trip	Flooded length (m)	Time increase (median, %)	No safe route (%)
Shortest time	16.76	1063.6	0.0	–
LR	12.25	756.1	5.4	94.7%
SVM (RBF)	10.65	653.1	9.2	94.7%
Random forest	10.69	655.7	9.7	94.7%
XGBoost	10.50	650.0	11.3	94.7%
MLP	**9.55**	**599.0**	20.3	94.7%
GraphSAGE	10.73	677.5	13.7	94.7%
Oracle (observed)	8.03	485.6	21.9	94.7%

Fig. 7. Flood exposure per trip versus median travel-time increase for each strategy at $\lambda = 5.0$.

emerged as the dominant features, followed by proximity to the Han River. During the October 2022 event, extreme rainfall coincided with high tidal river stages, inducing backwater ponding and disabling gravity drainage outfalls along the riverfront. The models accurately identified low-lying arterial underpasses (e.g., Dien Bien Phu) and riverbank road segments as high-risk clusters.

C. Cost of Model Complexity

Comparing model complexity against operational performance (Tables III and VI), MLP achieved the best balance of predictive accuracy and computation, requiring only 3,329 parameters and 1.6 ms for full district inference. GraphSAGE demonstrated the predictive value of topological message passing (+0.021 AUC-PR over the edge-free baseline) with an inference time of 8.3 ms. Given that all models evaluated completed district-wide inference well within 100 ms, the computational cost of neural and graph architectures is negligible compared to their gains in flood avoidance during routing.

D. Limitations

Five limitations apply. First, the study uses one event, so it identifies flood-prone segments rather than forecasting floods for any storm. Second, unreported flooding outside the excluded buffer is still labeled as non-flooded, which lowers the measured recall. Third, rainfall varies little across a single district, so it cannot separate segments within one event. Fourth, a coarse DEM may miss street-scale depressions. Fifth, crowd-sourced flood observations may introduce reporting bias toward major arterial corridors over smaller residential alleyways.

VII. CONCLUSION

This paper benchmarked six machine learning models from five families for road-level flood prediction and flood-aware routing during the 2022 Da Nang flood. Multilayer perceptron (MLP) reached the highest AUC-PR of 0.891 ± 0.042 under spatial cross-validation, and as a routing cost it reduced flood exposure by 43.7% for a 20.3% increase in travel time. All models executed district-wide inference in under 98 ms, and their spatial rankings proved robust across block scales from 250 m to 1000 m.

Future work will add further events, timestamped depth records for short-term prediction, and risk costs in reinforcement-learning routing [17]. The risk maps could also provide prior flood probabilities for placing road sensors or cameras in Da Nang, following [13].

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Fig. 8. Trade-off between flood exposure and travel-time increase as risk aversion λ varies from 1 to 20.

Fig. 9. Routes for a representative OD pair: shortest time (red), flood-aware ML detour (blue), and oracle (green).

TABLE V
METRICS OF THE REPRESENTATIVE ROUTE

Strategy	Travel time (min)	Flooded segments	Flooded length (m)
Shortest time	3.08	14	951.0
XGBoost	4.93	2	108.4
MLP	4.93	2	108.4
Oracle (observed)	5.08	0	0.0

TABLE VI
MODEL SIZE AND RUN TIME

Model	Parameters	Training time (s)	Inference, district (ms)	AUC-PR
LR	19	0.01	12.6	0.840
SVM (RBF)	37,062	0.34	97.4	0.856
RF	50,000	0.27	45.5	0.874
XGBoost	3,200	0.10	7.0	0.847
MLP	3,329	0.34	1.6	**0.891**
GraphSAGE	3,297	0.52	8.3	0.867

Routing time per OD pair: 4.2 ms for all models.

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Fig. 10. Sensitivity of the results to (a) spatial block size and (b) handling of uncertain segments.

TABLE VII
SENSITIVITY OF THE BEST MODEL TO VALIDATION AND LABELING SETTINGS

Setting	AUC-PR	Recall	Brier ↓
Block size 250 m	0.883	0.839	0.131
Block size 500 m (main)	0.847	0.839	0.131
Block size 1000 m	0.853	0.839	0.131
Uncertain segments excluded	0.987	0.955	0.053
Uncertain segments as non-flooded (main)	0.847	0.839	0.131
Broad flood ($\geq$10 cm)	0.986	0.903	0.087